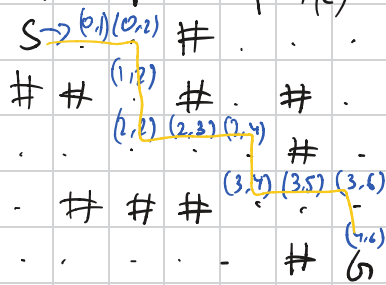


AI HW1 Rough Work

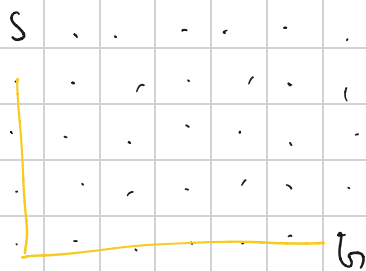
Syed Ibad Asim

1) First comparison part (c)



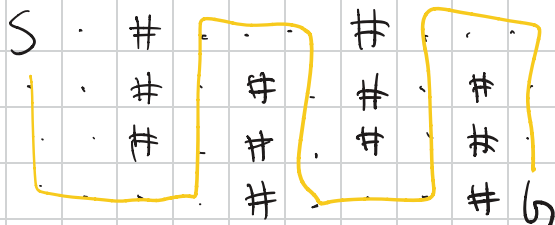
Cost 10 A* 12 Dijkstra 24

2) Second comparison open grid



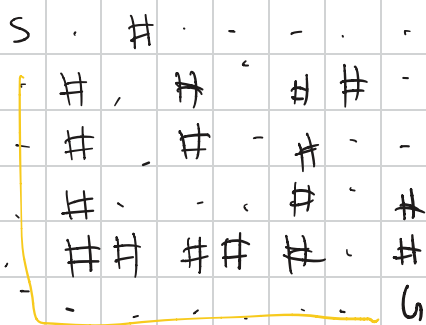
Cost 10 A* 11
Dijkstra 35

3) Comb



Cost 24 28 A* nodes
28 Dijkstra nodes

4) Maze



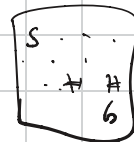
Cost 12 A* 13 nodes
Dijkstra 15 nodes

5) Example grid reversed



Cost 10 A* 17
Dijkstra 25

part a) First we need to define the movement, and the grid so we can establish our search.



← we'd also need a way to know where S and G are so define that too.

Make a class to define all possible actions. This should make our algorithm work.

- State representation - Every possible place on the grid (row, column) represents one state.

- Initial state - S represents the starting node.

- Goal state - G represents the last node.

- Actions - Up, down, left, right accounts for 1 cost value each.

- Successor functions - Based on the current cell try all 4 moves unless next place is a #. This way each move returns successor, action, cost.

- Cost: Each move costs 1.

- Heuristic Function for A*: Manhattan distance

$$h(r, c) = |r - r_{goal}| + |c - c_{goal}|$$

- If we apply this Manhattan distance formulae to the example we get.

$$|4 - 0| + |6 - 0| = 10$$

• Manhattan works as a solid heuristic for A^* because it never overestimates, and it's consistent